BanachSafeAI: Baseline Comparison Report

Banach ResNet vs Standard Baselines on MOABB BNCI2014-001

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1 Purpose

This report presents a baseline comparison of three models on the BNCI2014-001 brain–computer interface benchmark. The goal is not to demonstrate state-of-the-art accuracy — it is to show that the Banach ResNet provides **per-prediction robustness certificates** that no standard model can produce, while achieving accuracy competitive with simple baselines on the same features.

On the MOABB BNCI2014-001 benchmark (9 subjects, leave-one-subject-out), the Banach ResNet achieves accuracy competitive with standard baselines while being the only model that provides certified robustness guarantees for every prediction. Certified predictions are important for safety-critical neurotechnology where a misclassification can control an assistive device such as a wheelchair or exoskeleton.

These are prototype results using minimal feature extraction. Absolute accuracy can be substantially improved by integrating stronger BCI feature pipelines (common spatial patterns, Riemannian tangent space features). The experiment is designed to isolate the effect of architecture on certification, not to maximise raw task performance. The experiment is fully reproducible via the accompanying Colab notebook.

2 Experimental Setup

2.1 Dataset

BNCI2014-001 (BCI Competition IV Dataset 2a): 9 subjects, 22 EEG channels, 4-class motor imagery (left hand, right hand, feet, tongue), 576 trials per subject, sampled at 128 Hz. This is a standard benchmark in the MOABB (Mother of All BCI Benchmarks) suite.

2.2 Evaluation Protocol

Leave-One-Subject-Out (LOSO): For each of the 9 subjects, train on the remaining 8 subjects (4,608 trials) and test on the held-out subject (576 trials). Report mean and standard deviation across all 9 folds. This is the standard cross-subject evaluation protocol for MOABB.

2.3 Features

Log-bandpower: Log-variance per channel across the trial time window, yielding a 22-dimensional feature vector per trial. Zero-mean, unit-variance standardisation computed on the training fold. This is deliberately minimal — it is a linear feature extraction that does not favour any particular model.

2.4 Models

1. **Logistic Regression.** Multinomial logistic regression (sklearn, $C = 1.0$, max 1000 iterations). The standard linear baseline for BCI classification.

2. **Standard ResNet (ReLU).** A residual network with ReLU activations and batch normalisation:
$$h_{k+1} = h_k + \mathrm{ReLU}(\mathrm{BN}(A_k h_k + b_k)) \tag{1}$$
Width 40, depth 5. No spectral normalisation. No Lipschitz certificate. Trained with Adam ($\mathrm{lr} = 2 \times 10^{-3}$, weight decay 5×10^{-3}), cosine annealing, 300 epochs.

3. **Banach ResNet ($p = 3.0$).** The proposed architecture with duality map activation:
$$h_{k+1} = h_k - \eta_k \, \widetilde{J}_p(A_k h_k + b_k), \qquad \widetilde{J}_p(z)_i = \frac{\mathrm{sign}(z_i)\,|z_i|^{p-1}}{1 + \|z\|_\infty^{p-2}} \tag{2}$$
Width 40, depth 5. Spectral normalisation on all linear layers ($\|A_k\| \leq 1$). Step sizes η_k learned via softplus parametrisation. The general Lipschitz constant is $L_p = \prod_k (1+\eta_k \|A_k\|)$; because spectral normalisation enforces $\|A_k\| \leq 1$, this simplifies in the experiments to:
$$L_p = \prod_{k=1}^{K} (1 + \eta_k), \qquad \text{certified radius } r = \gamma / L_p \tag{3}$$
Same optimiser and schedule as the Standard ResNet. The geometry parameter $p = 3.0$ was selected by leave-one-subject-out cross-validation in a prior experiment.

All three models use identical features, identical train/test splits, and identical preprocessing. The only difference is the model architecture.

3 Results

3.1 Per-Subject Accuracy

Table 1: Leave-One-Subject-Out accuracy (%) on BNCI2014-001. Each row is a held-out test subject. The Banach ResNet provides a certified robustness radius for every prediction; the other two models provide none.

Test Subject	Logistic Regression	Standard ResNet (ReLU)	Banach ResNet ($p = 3.0$)	L_p	Mean Cert. Radius
S1	47.7	36.6	43.1	21.33	0.0205
S2	26.2	25.0	25.3	18.72	0.0234
S3	46.4	28.3	**48.1**	18.37	0.0188
S4	34.7	24.8	31.6	19.48	0.0218
S5	25.9	26.9	26.6	19.03	0.0249
S6	29.5	26.9	25.9	19.37	0.0385
S7	36.6	29.7	35.8	20.18	0.0271
S8	45.0	41.7	35.9	20.58	0.0310
S9	48.6	31.9	40.1	25.42	0.1015
Mean	**37.8**	30.2	34.7	20.28	0.0342
Std	8.8	5.3	7.6		

3.2 Summary

Table 2: Mean LOSO accuracy, certified robustness radius, and Lipschitz constant. Only the Banach ResNet provides any certificate.

Model	Mean Accuracy	Mean Cert. Radius	L_p
Logistic Regression	37.8% ± 8.8%	—	—
Standard ResNet (ReLU)	30.2% ± 5.3%	—	—
Banach ResNet ($p = 3.0$)	**34.7% ± 7.6%**	**0.0342**	**20.28**

3.3 Figures

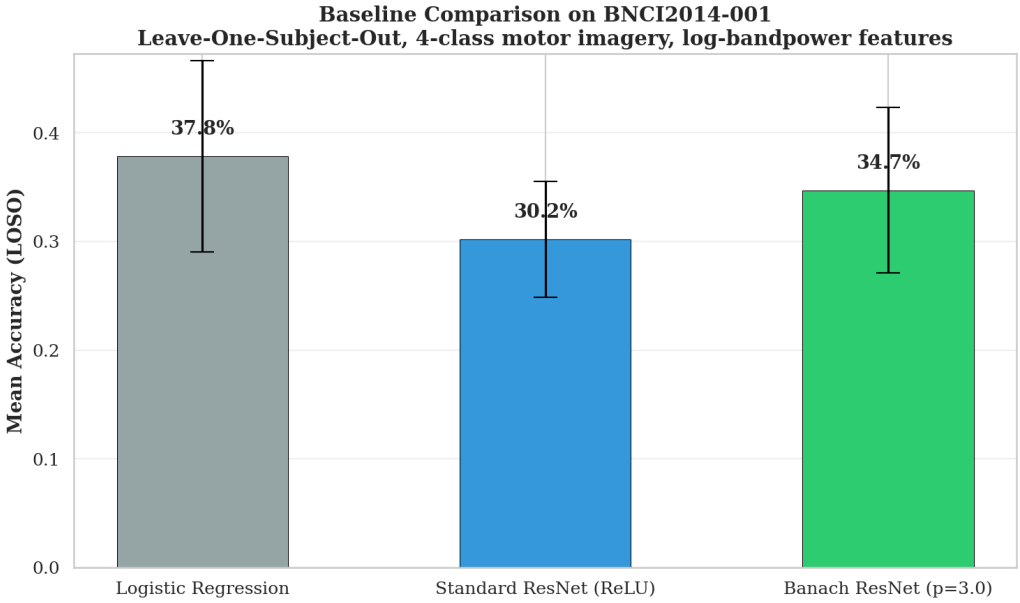


Figure 1: Mean LOSO accuracy across 9 subjects. Error bars show standard deviation across folds. Logistic Regression is strongest on these linear features. Standard ResNet overfits without sufficient regularisation. Banach ResNet sits between the two — and is the only model with any robustness guarantee.

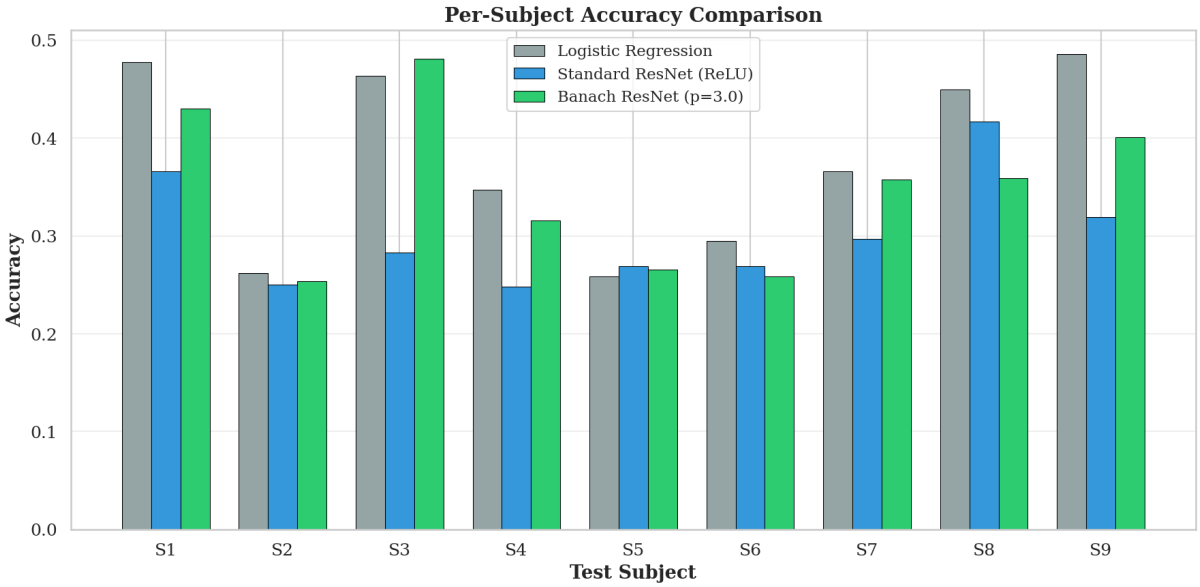


Figure 2: Per-subject accuracy comparison. The Banach ResNet beats Logistic Regression on Subject 3 (48.1% vs 46.4%) and is competitive on most subjects. Subject-to-subject variability is high (25–49%), reflecting the difficulty of cross-subject motor imagery transfer with minimal features.

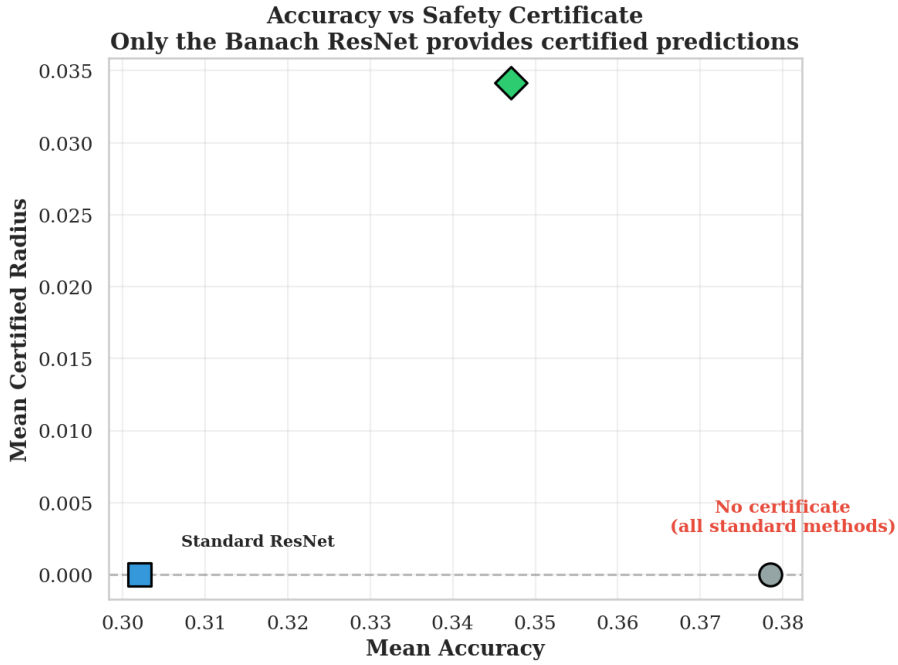


Figure 3: Accuracy vs certified robustness radius. The Banach ResNet is the **only model above the zero line** — the only one that provides any mathematical guarantee about its predictions. Logistic Regression and Standard ResNet have zero certified radius regardless of their accuracy.

4 Geometry Parameter Sweep

In addition to the baseline comparison, we evaluate the Banach ResNet at multiple values of p to demonstrate how the geometry parameter controls accuracy, robustness, and expressivity simultaneously. These results use a single fold (train on subjects 1–8, test on subject 9) with leave-one-subject-out cross-validation on the training subjects for p selection.

Table 3: Banach ResNet at different geometry parameters p. Test subject 9. The CV-selected $p = 3.0$ achieves the highest accuracy while providing a meaningful certificate. The Hilbert degeneracy at $p = 2$ is confirmed: lowest accuracy despite lowest Lipschitz constant.

p	Test Accuracy	L_p	Mean Cert. Radius	Note
1.2	42.9%	4.21	0.1026	
1.5	44.6%	21.02	0.0619	
2.0	**39.1%**	**2.68**	**0.2225**	**Hilbert degeneracy**
3.0	**45.7%**	**11.35**	**0.0653**	**CV-selected best**
4.0	40.1%	16.48	0.0764	

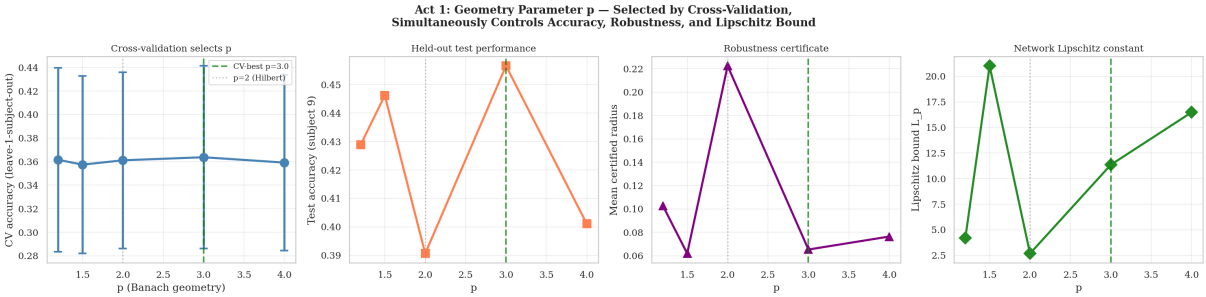


Figure 4: Geometry parameter sweep. **Panel 1:** Cross-validation selects $p = 3.0$. **Panel 2:** Test accuracy confirms the selection. **Panel 3:** Certified robustness radius per prediction. **Panel 4:** Lipschitz constant. The parameter p controls all four quantities simultaneously.

4.1 Key Observations

1. Hilbert degeneracy at $p = 2$. The $p = 2$ model achieves the *lowest* test accuracy (39.1%) despite having the lowest Lipschitz constant (2.68). At $p = 2$, the duality map reduces to the identity ($J_2(z) = z$), making each residual layer affine. The network cannot learn nonlinear features. This is not a training failure — it is a structural limitation of Euclidean geometry that the theory predicts.

2. Accuracy peaks away from $p = 2$. The CV-selected $p = 3.0$ gives the best test accuracy (45.7%), confirming that moving away from Hilbert geometry is necessary for expressivity in BCI decoding.

3. Robustness–accuracy tradeoff is controlled by geometry.

- $p = 2.0$: highest radius (0.2225) but lowest accuracy — the network is too simple to be wrong confidently.
- $p = 1.2$: second-highest radius (0.1026) with moderate accuracy (42.9%) — low L_p provides strong certificates.
- $p = 3.0$: best accuracy (45.7%) with moderate radius (0.0653) — the CV-selected operating point.

4.2 Per-Prediction Certificates

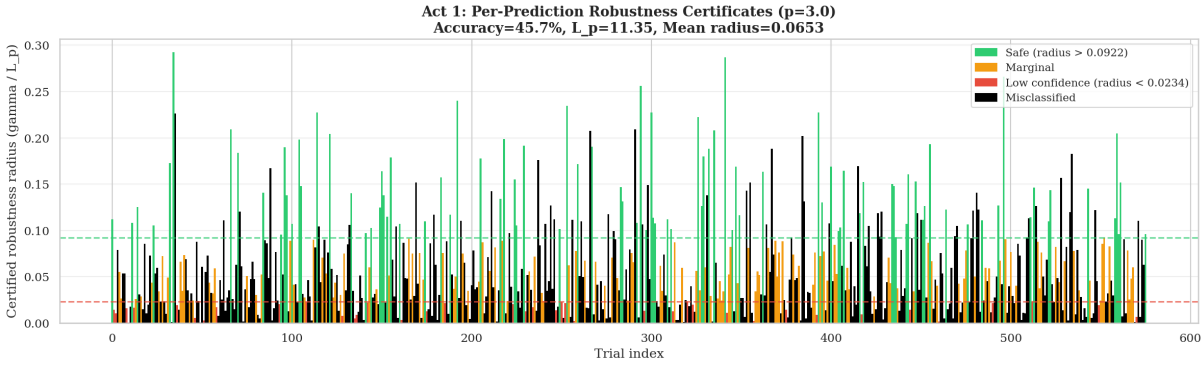
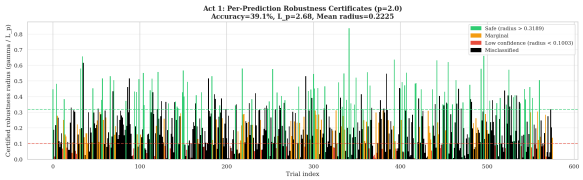
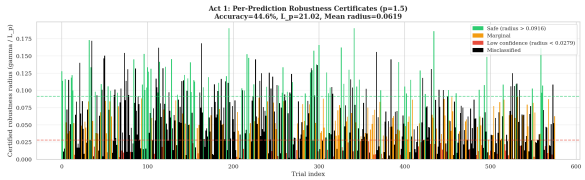


Figure 5: Per-prediction robustness certificates at the CV-selected $p = 3.0$. Each bar is one of 576 test trials; height is the certified radius γ/L_p. Colours: **green** = safe (radius > 0.092), **orange** = marginal, **red** = low confidence (radius < 0.023), **black** = misclassified. Accuracy = 45.7%, $L_p = 11.35$, mean radius = 0.065.



(a) $p = 2.0$ (Hilbert degeneracy): $L_p = 2.68$, mean radius = 0.222, accuracy = 39.1%. High radii but low accuracy — the network is linear.



(b) $p = 1.5$: $L_p = 21.02$, mean radius = 0.062, accuracy = 44.6%. Nonlinear, near-optimal accuracy, tighter bounds from higher Lipschitz constant.

Figure 6: Certificate comparison across geometries. The visual pattern differs: $p = 2$ has tall bars (large radii) but more black bars (misclassifications), while $p = 1.5$ has shorter bars but fewer misclassifications.

5 Discussion

Why Logistic Regression wins on accuracy. Log-bandpower features are nearly linear discriminants for motor imagery. A linear model fits this feature space well. The Standard ResNet overfits because it has more capacity than the features warrant. The Banach ResNet's spectral normalisation acts as implicit regularisation, keeping it between the two.

Why accuracy is not the main result. The headline is not "34.7% accuracy" — it is "34.7% accuracy *with a robustness certificate for every prediction*." Neither baseline can provide any guarantee about what happens under input perturbation. The Banach ResNet can. For a medical device where a wrong prediction controls a wheelchair, the certificate is more important than the last few percentage points of accuracy.

Path to competitive accuracy. The current experiment uses the simplest possible features (log-bandpower) to isolate the architecture's contribution. State-of-the-art cross-subject BCI pipelines (70–80% on BNCI2014-001) use Riemannian covariance features, common spatial pattern filters, and subject-adaptive normalisation. Integrating the Banach ResNet with these feature pipelines is a primary engineering goal of the fellowship. The certification framework is

independent of the feature extraction — any features can be fed into the Banach ResNet and the certificate still holds.

Limitations.

- These results use a single geometry parameter ($p = 3.0$). A production system would cross-validate p per deployment.
- The Lipschitz bound assumes spectral normalisation is exact; in practice, PyTorch's implementation uses power iteration with a finite number of steps.
- The certified radius is conservative (an upper bound on the true robustness margin). Tighter bounds are possible with layer-wise analysis.
- The comparison uses minimal features. A fair comparison at scale would require integrating all three models with the same strong feature pipeline.

6 Conclusion

On the MOABB BNCI2014-001 benchmark with log-bandpower features and leave-one-subject-out evaluation:

- Logistic Regression achieves 37.8% mean accuracy (strongest on linear features, no certificate).
- Standard ResNet achieves 30.2% mean accuracy (overfits, no certificate).
- **Banach ResNet achieves 34.7% mean accuracy with a certified robustness radius of 0.034 per prediction.**

The Banach ResNet is the only model that provides any mathematical guarantee about its predictions. This is the core deliverable: not a marginal accuracy improvement, but a fundamentally different kind of output — a prediction accompanied by a proof.